

Introduction

The International Telecommunication Union (2025) reported that 74% of the world's population in 2025, amounting to 6 billion people, were online as compared to 71% in 2024. In the Asia-Pacific region, internet usage rate stood at 77% (ITU, 2025a). Roughly 82% of youth aged 15-24 years used the internet globally (ITU, 2025b).

With technology advancing at an accelerated rate, internet use has increased multi-fold. The Internet has entrenched itself in all parts of life, be it in schools or universities, at work, or in hospitals. Globally, people use the internet for various purposes such as browsing for information, communication, and entertainment. Currently, a very large proportion of internet activity is related to information seeking and finding answers when compared to 22 years ago (Liu, 2020; Wellman et al., n.d).

The internet has become a primary source of health-related information. Several young adults aged 18-30 years turn towards the internet for health-related searches (Stifjell et al., 2025). In 2015, every one in twenty Google searches was related to health-related information. The term 'Dr Google' refers to searching for information pertaining to symptom identification and diagnoses of illnesses using Google search engine used by both, patients as well as physicians (Aslantas & Altuntas, 2023; Ktak et al., 2017). Eysenbach and Kohler (2003) found that approximately 4.5% of searches on the World Wide Web were health-related with a global estimate of 6.75 million health-related searches every day. An international study across 12 countries and 12,000 participants revealed that at least three in five respondents used the internet to access medical related information with information related to medicines being the most frequently searched (McDaid & Park, 2010).

While health-related online searches can help people make informed healthcare decisions, some users spend excessive time on such internet searches. Exposure to excessive health information can cause distress and anxiety especially in individuals with limited medical knowledge. Difficulty in reading and understanding medical information increases ambiguity. People with low tolerance of uncertainty may assume that they have the most serious medical conditions, thereby increasing anxiety and continued internet searches. On the other hand, people with higher tolerance of uncertainty will be more successful in reassuring themselves that they do not have an illness and stop after one or two internet searches (Blackburn et al., 2019).

Hypochondriasis or health anxiety, currently known as Illness Anxiety, refers to “extreme anxiety and the distress that an individual experiences because of inaccurate interpretations of his or her body sensations” (Nadeem et al., 2022, p.1). People with health anxiety and chronic pain may engage in certain safety behaviours to alleviate anxiety and distress associated with the belief that they have a severe disease. Such behaviours may include reassurance seeking through repeated online searches for health-related information (Nadeem et al., 2022).

Cyberchondria is the portmanteau of the words cyber and hypochondriasis suggesting that it is a form of hypochondriasis that is caused by internet use (Starcevic & Berle, 2013; Zheng et al., 2021). It refers to the “excessive or repeated search for health-related information on the Internet, driven by distress or anxiety about health, which only amplifies such distress or anxiety” (Starcevic & Berle, 2013, p.206). Individuals with cyberchondria experience a misguided apprehension of serious medical illness based on reviewing information available on the web. Cyberchondria escalation has been linked to the availability

of health information, presence of escalatory terminologies in pages visited by the respondents as well as the respondent's predisposition to mounting anxiety versus seeking more rational explanations for ailments (White & Horvitz, 2009).

Despite being correlated, health anxiety and cyberchondria are different. Health anxiety refers to disproportionate, excessive concerns related to physical conditions. On the other hand, cyberchondria exhibits characteristics beyond worry and anxiety that are caused by viewing medical information that may be threatening in nature (Zheng et al., 2021).

Other psychological variables also influence the development and maintenance of cyberchondria including metacognitions and generalized anxiety. Metacognition can be defined as “psychological structures, knowledge, events and processes that are involved in the control, modification and interpretation of thinking itself” (Wells & Cartwright-Hatton, 2004, p. 386). Elevated maladaptive metacognitive beliefs are seen in individuals with various psychological disorders and physical illnesses (Keen et al., 2022). Cognitive models attribute health anxiety to misinterpretations of symptoms whereas the metacognitive model, grounded in the Self-Regulatory Executive Function (S-REF) model, attributes health anxiety to difficulty in regulating worry about symptoms rather than seeing it as a problem of believing one is ill (Bailey & Wells, 2015). Metacognitive beliefs lead people to believe that thinking about illnesses keeps them safe, making them more attentive towards their health as well as contributing to further health-related searches on the internet (Nadeem et al., 2022).

General anxiety has been included in this study as a psychological variable that may influence how individuals respond to uncertainty and threatening health-related information. People with anxiety are very likely to search for information online to reduce their anxiety. However, due to the vast amount of information available on the web, their apprehensions

increase thereby increasing their anxiety (Batool et al., 2022). A few studies have demonstrated the relationship between cyberchondria and general anxiety (Gioia & Boursier, 2020; Han et al., 2021; Schenkel et al., 2021).

To summarise, cyberchondria, characterized by excessive online search for health-related information, is triggered by the very anxiety which it hopes to reduce. It may actually have an opposite, anxiety-inducing effect depending on the search outcome. Additionally, metacognitions (positive metacognitions such as reduction of uncertainty or negative metacognitions such as fear of loss of control) play a dominant role (Jungmann et al., 2024).

In the Indian context, empirical studies on cyberchondria are still emerging. Several studies have looked at prevalence of cyberchondria in Indian adults (Kulsum et al., 2023; Makarla et al., 2019; Vemuri & Prusty, 2025). Cyberchondria has been correlated with various variables such as smartphone addiction (Agrawal et al., 2024), general mental health (Makarla et al., 2019), and depression (Shrivastava et al., 2022).

Rationale for Study

The Cognitive-Behavioural model of cyberchondria proposes that people search for online health-related information as a form of reassurance seeking. However, instead of reducing anxiety, it may show worsening in some individuals (Makarla et al., 2019). As per the metacognitive model of cyberchondria, positive metacognitive beliefs (advantages of health-related worry) and negative metacognitive beliefs (uncontrollability of health-related worries) result in persisting health-related worry resulting in greater distress (Bailey & Wells, 2015). Hence, individuals continue searching for health-related information online, not just because of anxiety, but also because of their beliefs on usefulness or pitfalls of their health-related worries. Studies have established a relationship between cyberchondria and anxiety.

However, it is also necessary to understand the mechanism underlying this relationship. The rationale of this study is to explore whether metacognitive beliefs mediate the relationship between cyberchondria and anxiety.

Research Aim and Objectives

The present study aims to understand the relationship between cyberchondria, metacognitive beliefs and anxiety on young Indian adults (18-30 years). The study aims to specifically investigate metacognitive beliefs as a potential mediator between cyberchondria and anxiety. Additionally, the study also aims to explore gender differences in cyberchondria levels.

The objectives of this study are:

1. To examine gender differences among young adults.
2. To assess the relationship between cyberchondria, metacognitive beliefs and anxiety.
3. To determine the predictive role of metacognitive beliefs and anxiety on cyberchondria.
4. To investigate the mediating effect of metacognitive beliefs on cyberchondria and anxiety.

Research Hypotheses

Hypothesis 1

H₀₁ - There is no significant difference in cyberchondria severity between men and women.

H₁ - Women will report significantly higher cyberchondria severity as compared to men.

Hypothesis 2

H0₂ - There are no significant correlations between cyberchondria, metacognitive beliefs, and general anxiety.

H₂ - Cyberchondria will be significantly positively correlated with metacognitive beliefs and general anxiety.

Hypothesis 3

H0₃ - Metacognitive beliefs will not explain greater variance in cyberchondria than general anxiety.

H₃ - Metacognitive beliefs will explain greater variance in cyberchondria better than general anxiety

Hypothesis 4

H0₄ - Metacognitive beliefs will not mediate the relationship between cyberchondria and general anxiety; general anxiety will remain a significant predictor after accounting for metacognitive beliefs.

H₄ - Metacognitive beliefs will fully mediate the relationship between cyberchondria and general anxiety.